

Project Report

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This report details the final-grade calculation methodology, the core security mechanism for enforcing roles and maintenance mode, and provides a clear list of tables used in both the Authentication (Auth) and ERP Main databases.

1. Final-Grade Weighting Rule

The University ERP system uses a flexible, weighted average model to determine the final grade for a student in a specific course section. Instructors define the assessment components, maximum scores, and weights.

1.1 Calculation

The final grade is calculated as a weighted percentage based on the component scores, where weights are assumed to be normalized across all components.(This calculation is performed in the GradeDao.
`calculateFinalGrade` method.)

1.2 Letter Grade Conversion Scale

The resulting numeric percentage is converted into a standard letter grade by the StudentService using the following fixed scale:

Numeric Grade	Letter Grade
$\geq 90\%$	A
$\geq 80\%$	B
$\geq 70\%$	C
$\geq 60\%$	D
$< 60\%$	F

2. Enforcement of Roles and Maintenance Mode

Access control is centralized in the application's service layer to ensure data integrity and system security, regardless of the UI entry point. The core enforcement logic resides in `edu.univ.erp.access.AccessRuleChecker`.

2.1 Role-Based Access Control

The `AccessRuleChecker.checkWriteAccess(roleRequired)` method is the primary gate for all actions that modify data (INSERT, UPDATE, DELETE). The roles are enforced hierarchically:

1. **Admin:** Any user attempting an action requiring the ADMIN role must be an Admin.
2. **Instructor Scope:** An INSTRUCTOR can perform actions requiring INSTRUCTOR or STUDENT roles.
3. **Student Scope:** A STUDENT can only perform actions requiring the STUDENT role.

The application also strictly enforces **Data Ownership** via `AccessRuleChecker.checkDataOwnership(requestedUserId)`, ensuring that non-Admin users (Students and Instructors) can only view or modify records associated with their own `user_id`.

2.2 Maintenance Mode Enforcement

Maintenance Mode (`settings.maintenance_on = 'true'`) is the highest priority access restriction. The enforcement flow is as follows:

1. **State Management:** The Admin toggles the mode in the `AdminService`, which updates the persistent state and immediately synchronizes the in-memory state within the `SessionManager`.
2. **Write Block:** When `SessionManager.isMaintenanceModeOn()` is true, any call to `checkWriteAccess()` immediately fails and returns "Access Denied: Maintenance Mode is ON."
3. **Exemption:** Only the ADMIN role is exempt from the Maintenance Mode write block.
4. **Visibility:** A persistent banner is displayed on the Student and Instructor dashboards, alerting users that the system is currently view-only.

2.3 Enhanced Security and Data Integrity Checks (Bonus)

The application implements several security and integrity features beyond the core requirements:

- **Login Lockout:** The `AuthenticationService` tracks failed login attempts. After 5 consecutive failures, the account is locked for 15 minutes (using `lockout_until` in the `users_auth` table).

- **Password Hashing Detail:** Passwords are secured using BCrypt with 10 log rounds, a robust industry standard.
- **Password History Check:** The changePassword function checks the dedicated password_history table to prevent users from reusing their last 3 passwords.
- **Section Deletion Block:** To maintain data integrity, the AdminService prevents the deletion of any section if it has any active or completed enrollments (REGISTERED or COMPLETED status).

3. Database Table Structures

The project uses two separate MySQL databases linked by the shared user_id.

3.1 Auth DB (erp_auth) - Credentials and Security

This database stores all user authentication and security metadata.

Table	Primary Key (PK)	Foreign Key (FK)	Critical Fields	Notes
users_auth	user_id	N/A	username, role, password_hash, status, failed_login_attempts, lockout_until	Primary record for login validation.
password_history	id	user_id → users_auth	password_hash	Stores previous hashes to prevent password reuse.
login_attempts	id	N/A	username, attempt_time, success	Tracks failed login attempts for audit and lockout enforcement.

3.2 ERP Main DB (erp_main) - Academic Data

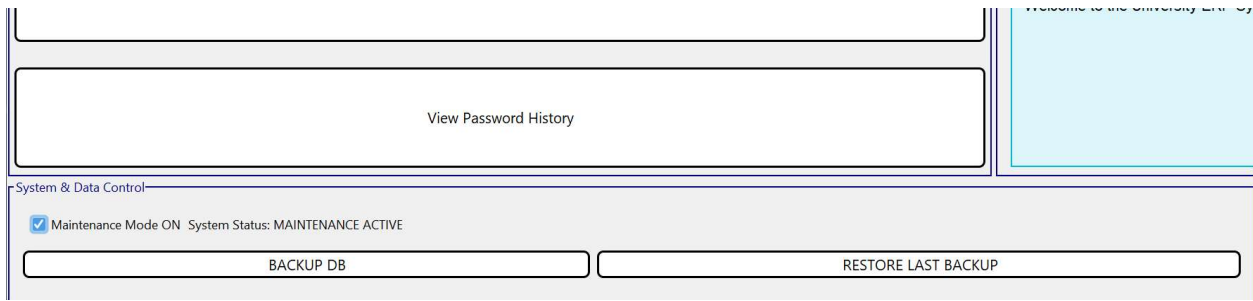
This database stores all academic and administrative application data.

Table	Primary Key (PK)	Foreign Key (FK)	Critical Fields	Notes
students	student_id (PK, FK)	N/A	roll_no, program, year	Profile data linked to users_auth.
instructors	instructor_id (PK, FK)	N/A	department, title	Profile data linked to users_auth.
courses	course_id	N/A	code, title, credits	Master course definitions.
sections	section_id	course_id, instructor_id	day_of_week, start_time, room, capacity, semester, year	Specific instance of a class offering.
enrollments	enrollment_id	student_id, section_id	status (REGISTERED/DROPPED/COMPLETED)	Links students to sections (prevents duplicates)

				via Unique Key).
grades	grade_id	enrollment_id	component, score, max_score, weight	Stores individual assessment scores.
settings	setting_key	N/A	setting_value (maintenance_on flag)	General application settings.
notifications	id (Fixed to 1)	posted_by_user_id	message, posted_by_role, posted_at	Stores the single global announcement.

Screenshots

1. Maintenance mode on



2. View only mode enabled

Student Dashboard - 2024246

SYSTEM IS IN MAINTENANCE MODE (VIEW ONLY)

Welcome, 2024246 (stu1) | Role: Student Logout

Navigation

Browse Course Catalog

View Grades & Transcri...

View Time Table

Change Password

My Registered Sections (Timetable View)

Section ID	Code	Course Title	Schedule	Room	Instructor
1	CSE101	Introduction to Prog...	Monday 09:00-10:30	C01	inst1
2	ECE111	Digital Circuits	Tuesday 11:00-12:30	C02	inst1
3	MTH100	Maths I (Linear Alge...	Wednesday 14:00-1...	C03	inst1

ANNOUNCEMENT by admin1 at Nov 2

3. Instructor graded enrolled student.

Grade Book for CSE101

Section: CSE101 (1) - Introduction to Programming

Enter Scores

Final Grade Calculation

Class Statistics for CSE101 (1)

Total Enrolled: 1

Average Final Grade: 83.00%

Component Averages %:

- Midterm: 90.00%

- Quiz: 80.00%

- EndSem: 80.00%

4. Student's grade transcript.

Transcript for 2024246						
Course	Title	Term	Status	Components	Final Grade	Letter
CSE101	Introduction to Progr...	S1 2024	REGISTERED	EndSem (80.0/100.0), Midterm (90....	83.00%	B
ECE111	Digital Circuits	S1 2024	REGISTERED	N/A	0.00%	N/A
MTH100	Maths I (Linear Algebr...	S1 2024	REGISTERED	N/A	0.00%	N/A

5. Password History window

Admin: View User Password History

Target User: Select or Search

Enter Username

Search

Close

Password Hash History

Index	Password ...	Length
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6. Notification Panel

Notification Panel

Message:

Welcome to the University ERP System!

PUBLISH

CLEAR

Live Notifications:

ANNOUNCEMENT by admin1 at Nov 2

Welcome to the University ERP Sys

7. Load and restore DB backup

System & Data Control

☐ Maintenance Mode ON System Status: Normal Operation

8. Change Password Option

Change Password

Change Password for: admin1

Current Password:

New Password (min 6 chars):

Confirm New Password:

9. Login Window

University ERP - Welcome

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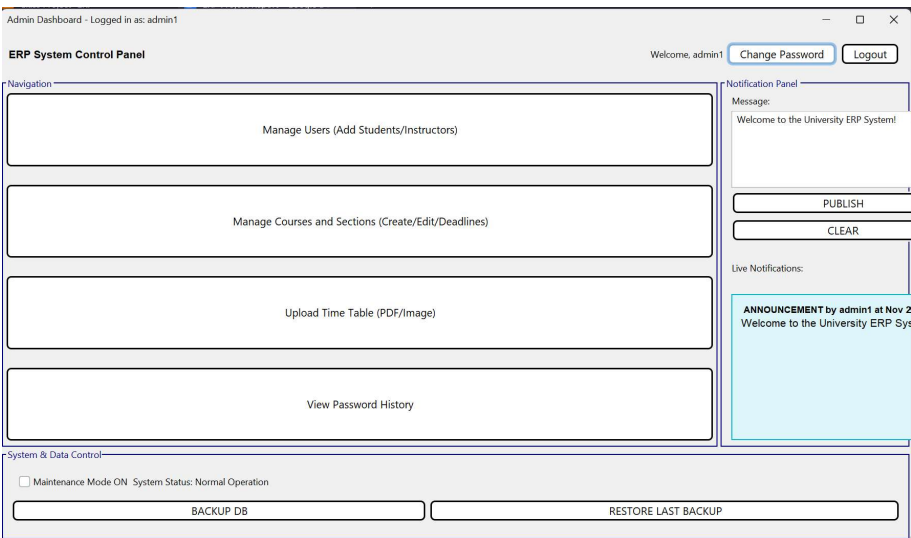
UNIVERSITY ERP

Please sign in using your credentials.

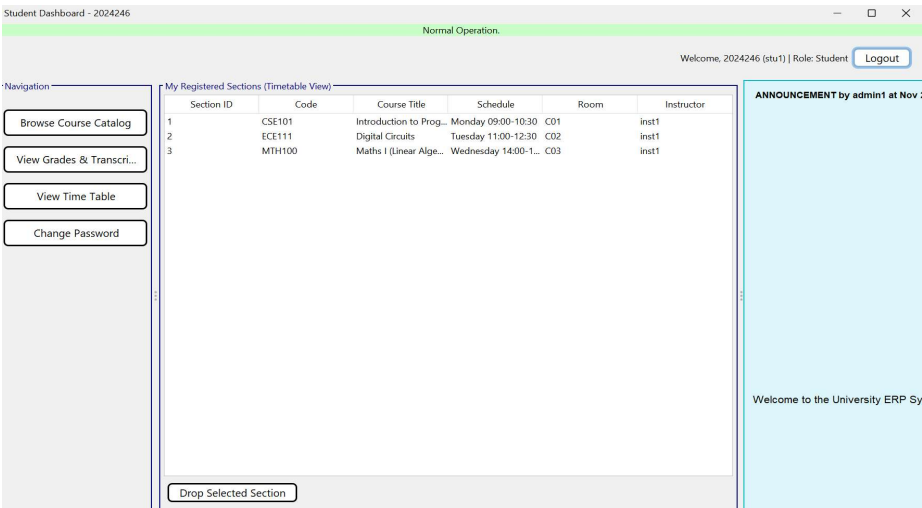
SIGN-IN

©University ERP System

10. Admin Dashboard



11. Student Dashboard



12. Instructor Dashboard

